Predictive and Statistical Analysis of Student Academic Performance Using Data Science

Data Science Research Capstone Report

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Domain: Education and Data Science
Dataset: UCI Student Performance — Mathematics
Target: Final Mathematics Grade (G3)

Abstract

This study applies statistical analysis and machine learning to investigate and predict student academic performance. The UCI Student Performance Mathematics dataset contains 395 observations and 33 variables. Statistical tests were used to examine the relationship between study time and final grade. A Linear Regression model was developed using 15 student-related features and achieved RMSE 2.1222, MAE 1.3482 and R^2 0.7804. PCA reduced 16 numerical variables to 11 components explaining 90.36% of variance. K-Means, DBSCAN and hierarchical clustering showed weak separation. A separate 30-day demand dataset was analyzed using SARIMA, producing test RMSE 0.5548 and MAPE 0.26%.

Keywords

Student performance; machine learning; statistics; Linear Regression; PCA; clustering; SARIMA.

1. Introduction

Student performance depends on academic, behavioral, demographic and social factors. Data science provides methods for identifying patterns, testing hypotheses and building predictive systems that can support educational decision-making.

Research Question

Can student demographic, academic, behavioral and study-related characteristics be used to understand and predict final mathematics grade (G3)?

Objectives

- Analyze factors associated with academic performance.
- Perform statistical hypothesis testing.
- Predict final mathematics grade.
- Apply PCA and clustering.
- Demonstrate time-series forecasting.

2. Literature Background

Educational data mining has been widely used to study student performance. Previous work indicates that demographic, behavioral and academic variables can provide useful information for understanding student outcomes. Statistical methods are useful for testing relationships and differences, while machine-learning methods can model multiple predictors simultaneously. Dimensionality reduction methods such as PCA can summarize correlated variables, and clustering can identify groups with similar characteristics.

This capstone combines these approaches into one reproducible data-science workflow. The main analysis uses the UCI Student Performance Mathematics dataset. The forecasting experiment uses a separate 30-day demand series to demonstrate time-series methodology.

3. Data and Methodology

The mathematics dataset contains 395 student observations and 33 attributes. The target variable G3 is the final mathematics grade on a 0–20 scale. Predictors include age, parental education, travel time, study time, failures, family relationships, free time, social activity, alcohol consumption, health, absences, G1 and G2.

Statistical analysis used a 0.05 significance level and included Shapiro-Wilk, Kolmogorov-Smirnov, Welch's t-test, Mann-Whitney U, one-way ANOVA, Tukey HSD, two-way ANOVA, Spearman correlation, Kruskal-Wallis and confidence intervals.

Predictive modeling used an 80/20 train-test split with random_state=42. Fifteen numerical features were standardized with StandardScaler before Linear Regression. RMSE, MAE and R^2 were used for evaluation.

4. Statistical Analysis Results

The Kruskal-Wallis test produced H = 7.5789 and p = 0.0556. This was slightly above the 0.05 threshold. The estimated epsilon-squared effect size was 0.0117.

Spearman correlation between study time and final grade was ρ = 0.1052 with p = 0.0367, indicating a statistically significant but weak positive association. Welch's t-test gave t = -2.1831, p = 0.0307, while Mann-Whitney U gave U = 11315.5, p = 0.0060.

One-way ANOVA gave F = 1.7278, p = 0.1607, and Tukey HSD did not show significant adjusted pairwise differences. Two-way ANOVA showed p = 0.027745 for study time, p = 0.004288 for sex, and p = 0.378450 for the interaction.

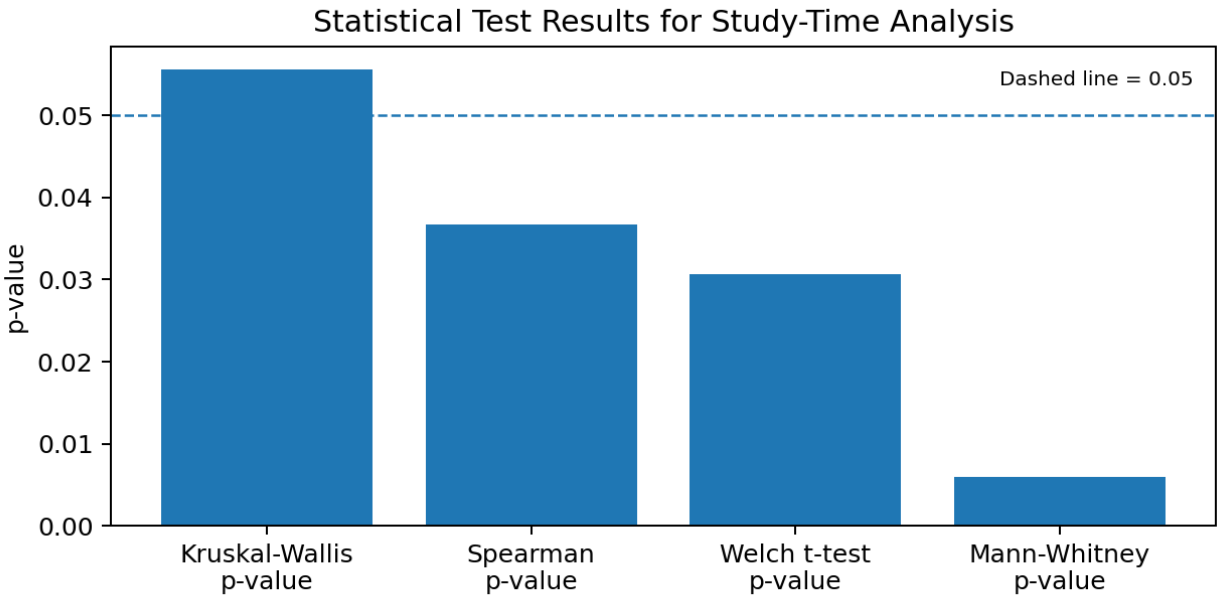


Figure 1. Reported p-values from the study-time statistical analyses.

Overall, study time showed only a weak association with final academic performance. This suggests that academic performance is influenced by multiple factors rather than study time alone.

5. Predictive Modeling Results

A Linear Regression model was developed using 15 student characteristics. The dataset was divided into 316 training observations and 79 testing observations. StandardScaler was fitted on the training data and then applied to both training and testing predictors.

Model performance on the test set was:

Metric	Result
RMSE	2.1222
MAE	1.3482
R^2	0.7804

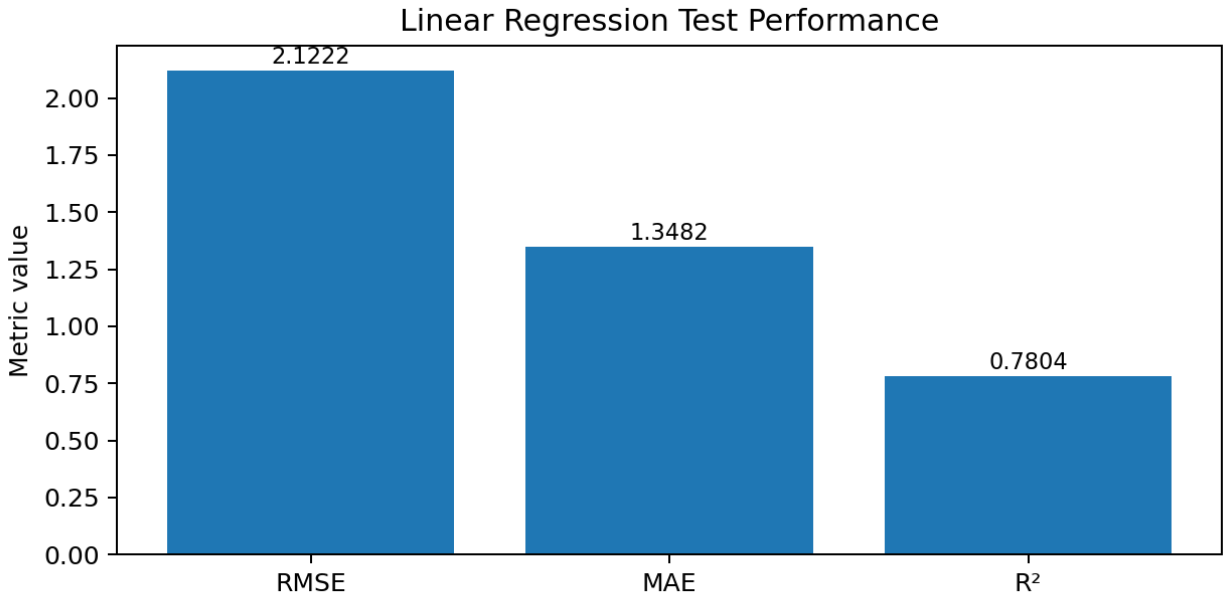


Figure 2. Linear Regression performance using the reported test metrics.

The model explained approximately 78% of the variation in final grades on the test data. G1 and G2 were important predictors because they represent earlier academic performance. Therefore, this model is most appropriate after earlier grades are available and should not be interpreted as a causal model.

6. Dimensionality Reduction and Unsupervised Learning

Sixteen numerical variables were standardized before PCA. The first 11 principal components explained 90.36% of the total variance. This provided a lower-dimensional representation while retaining most of the information in the selected numerical variables.

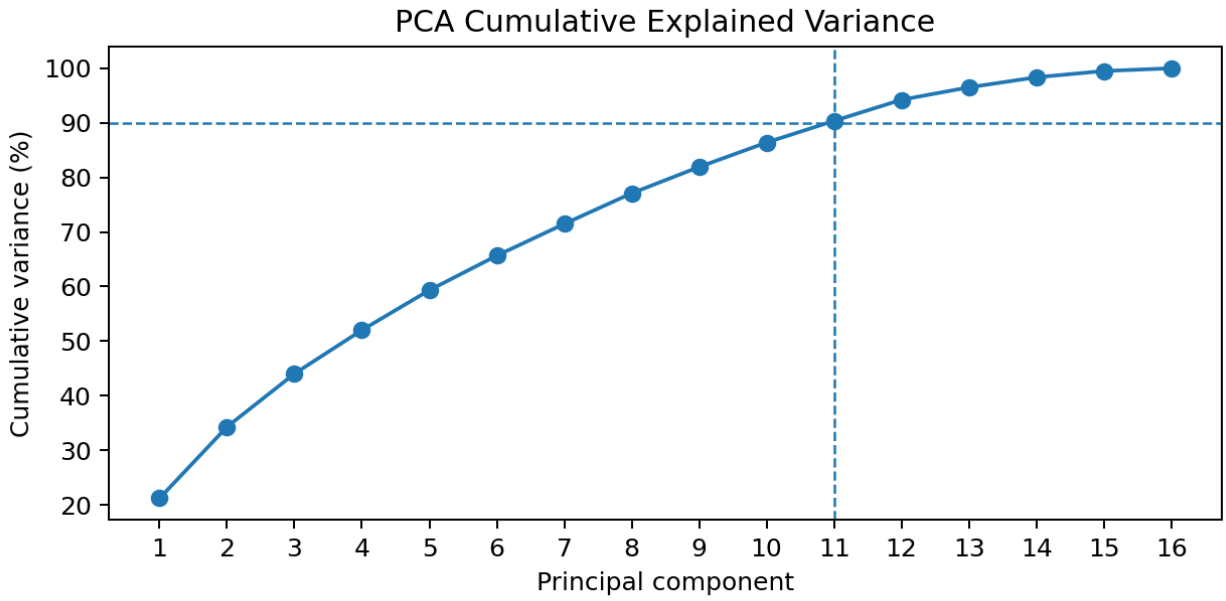


Figure 3. Cumulative explained variance from the 16-component PCA.

K-Means clustering with two clusters produced a silhouette score of approximately 0.1477. Agglomerative clustering produced approximately 0.0932. DBSCAN with eps = 3.5 produced one main cluster and approximately 50 noise observations.

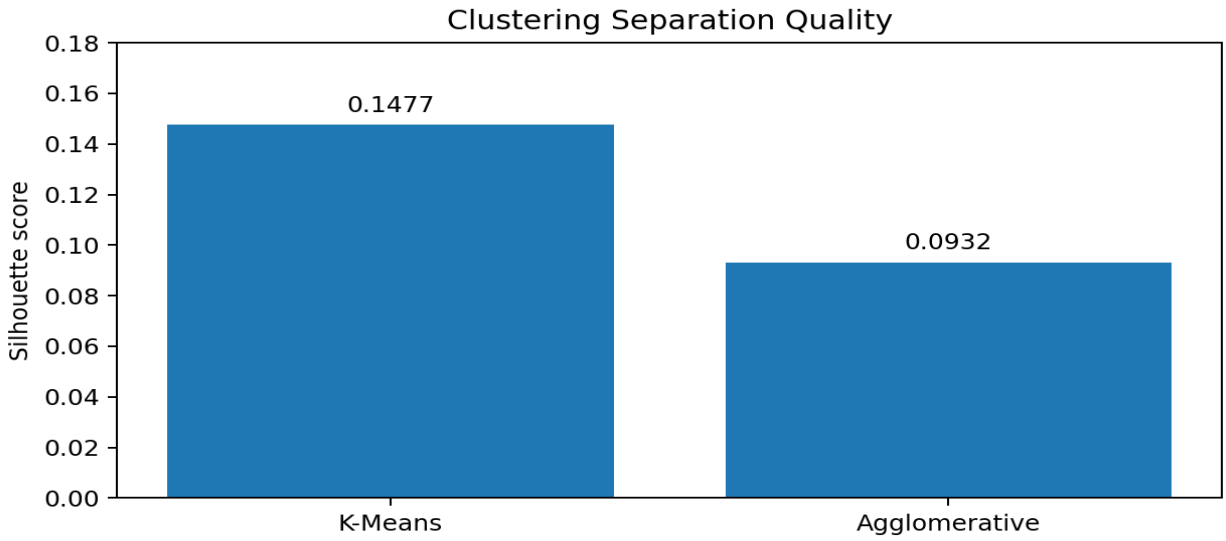


Figure 4. Silhouette scores for the clustering methods where applicable.

The low silhouette values indicate weak separation between student groups. Clustering therefore provides limited evidence of clearly distinct student profiles in the selected feature space.

7. Time-Series Forecasting

A separate 30-day demand dataset was used to demonstrate time-series forecasting. The series was decomposed using a weekly seasonal period. The original ADF test gave statistic -0.9039 and p = 0.7867, indicating non-stationarity. First differencing produced statistic -2.236 and p = 0.1943, so the series still did not meet the stationarity threshold.

Candidate ARIMA/SARIMA configurations were compared using AIC. The selected model was SARIMA(0,1,1) × (0,1,0,7), with AIC = 64.3152.

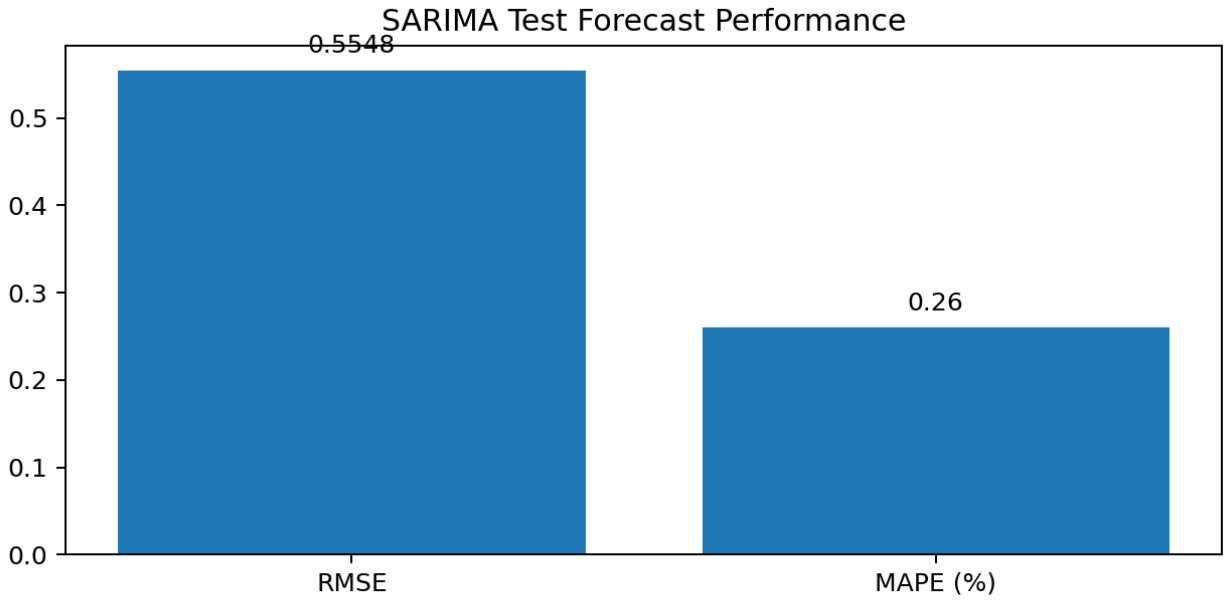


Figure 5. SARIMA test performance using the reported RMSE and MAPE.

The test RMSE was 0.5548 and MAPE was 0.26%. The model was subsequently refitted on the complete 30-day series to produce a 30-day forecast. Because this is a short demonstration series, the forecast should be interpreted cautiously.

8. Educational and Business Implications

The predictive analysis can support early identification of students who may benefit from academic support. Schools or educational platforms could use validated models as decision-support tools to prioritize tutoring, monitor progress and design personalized interventions.

The analysis also demonstrates an operational forecasting workflow. Similar methods can be applied to attendance demand, resource usage, enrollment-related demand or other educational operations when sufficiently long time-series data are available.

Responsible Use

Predictive models should support educators rather than replace human judgment. Student data should be handled with appropriate privacy, fairness and transparency safeguards. Model outputs should not be treated as definitive labels about student ability.

The results in this study describe associations and predictions. They do not establish that changing study time or any other variable will directly cause a change in final grade.

Practical Uses

- Early academic-support screening.
- Personalized learning recommendations.
- Monitoring of student progress.
- Data-informed institutional planning.
- Operational demand forecasting.

9. Limitations and Future Work

The main student-performance dataset is relatively small, containing 395 observations. It also represents a particular educational context and therefore may not generalize to all institutions or student populations.

The clustering experiments produced low silhouette scores, indicating that the selected numerical features do not naturally form strongly separated groups. The forecasting experiment used only 30 days of data, which limits the reliability of seasonal and long-horizon conclusions.

Another important limitation is the inclusion of G1 and G2 in the predictive model. These earlier grades are strong predictors of G3, but they are only available after earlier assessments. Consequently, the model is not a pure pre-assessment early-warning system.

Future Work

Future research could use larger and longitudinal datasets, compare Random Forest and Gradient Boosting models, perform feature engineering, evaluate cross-validation stability, and apply explainable AI methods. For forecasting, longer time-series datasets with multiple seasonal cycles and external predictors would provide stronger evidence.

10. Conclusion

This study demonstrated a complete data science workflow for student academic performance. Statistical analysis found a weak association between study time and final grade. The Linear Regression model achieved $R^2 = 0.7804$, showing useful predictive performance on the held-out test set.

PCA retained 90.36% of variance using 11 components, while clustering showed weak separation between student groups. The SARIMA forecasting experiment achieved test RMSE = 0.5548 and MAPE = 0.26% on the separate demand dataset.

Overall, combining statistical analysis, predictive modeling, dimensionality reduction, clustering and forecasting provides a useful framework for data-driven research. The findings should be interpreted as statistical associations and predictive results rather than causal conclusions.

Reproducibility

The accompanying project repository contains the analysis notebooks, source code, raw datasets, preregistration, requirements and supporting report files. The workflow was developed in Python using pandas, NumPy, Matplotlib, scikit-learn, statsmodels and Jupyter.

11. References

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Appendix: Key Results

Study-time analysis: $H = 7.5789$, $p = 0.0556$; Spearman $\rho = 0.1052$, $p = 0.0367$; $\varepsilon^2 = 0.0117$.
Predictive model: RMSE = 2.1222; MAE = 1.3482; $R^2 = 0.7804$.
PCA: 11 components, 90.36% cumulative variance.
K-Means silhouette ≈ 0.1477; Agglomerative silhouette ≈ 0.0932.
SARIMA: $(0,1,1) \times (0,1,0,7)$; AIC = 64.3152; RMSE = 0.5548; MAPE = 0.26%.